

Q. Discuss the Projection operation in Relational Algebra. Describe its syntax and explain its behavior regarding duplicate elimination, using suitable examples.

 **Definition to Remember**

The **Projection Operation** (denoted by π) is a **unary operation** in relational algebra that filters specific **columns (attributes)** from a relation while discarding the rest. It performs **vertical filtering** and automatically **eliminates duplicate rows** in the result. It is equivalent to the `SELECT column_name` in SQL.

1. Projection Operator (π — Pi)

- Operates on a **single table** (unary).
- Filters **vertically** — returns all rows, but only selected columns.
- Key Property:** Automatically **eliminates duplicate rows** from the output (because a relation is a mathematical set — no duplicates allowed).

2. Syntax of Projection

$\pi_{\{ \text{attribute_list} \}}(\text{Relation_Name})$

- attribute_list:** Comma-separated column names to retain.

3. Examples

Input Relation: STUDENT

Roll_No	Name	Major	City
101	John	CS	New York
102	Emma	Math	Boston
103	Luke	CS	New York
104	Mia	Physics	Chicago

Example 1: Simple Projection

Query: Get names and majors of all students.

$\pi_{\{ \text{Name, Major} \}}(\text{STUDENT})$

Result: 4 rows, 2 columns (Name, Major).

Example 2: Duplicate Elimination

Query: List all unique majors.

$\pi_{\{ \text{Major} \}}(\text{STUDENT})$

Result: CS, Math, Physics — *only 3 rows* (CS appeared twice in the original table but appears once in the result after duplicate elimination).

Example 3: Composition — Selection + Projection

Query: Find names of CS major students.

$\pi_{\{Name\}}(\sigma_{\{Major = 'CS'\}}(STUDENT))$

Execution:

1. **Step 1:** $\sigma_{\{Major='CS'\}}$ filters rows → John and Luke.
2. **Step 2:** $\pi_{\{Name\}}$ extracts the Name column → John, Luke.

★ Must-Write Points (for 10 marks)

1. Projection is a **unary operation** — operates on one relation.
2. Denoted by **π (Pi)**; equivalent to SQL's `SELECT column_name`.
3. Performs **vertical filtering** — selects specific columns, discards others.
4. Critical feature: **automatically eliminates duplicate rows** from the result.
5. The number of columns in output = number of columns listed in the attribute list.
6. Can be **composed with Selection** — Selection applied first, Projection second.
7. Duplicate elimination is automatic because a relation is a mathematical set.

⚡ Quick Recall

π (Pi) → Unary → Vertical Filtering (columns) → Duplicate Elimination (set) → Equivalent to SQL `SELECT` → Composable with σ